Curriculum Vitae

Tokey Sifullah Tanjil



Personal Information:

Name:	Tokey Sifullah Tanjil
Current Position:	Research Assistant, CUET (Dec 2025 to Present)
Location:	Chattogram, Bangladesh
Citizenship:	Bangladeshi
E-mail:	tsifullaht@gmail.com
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ORCID:	orcid.org/0009-0008-6421-7897
Languages:	English (fluent working proficiency), Bangla (native)

Professional Experience:

- **Research Assistant at CUET** *From 12 December, 2025 – Present*

Currently Working as a research assistant under the supervision of Prof. Dr. M.G. Hafez, Chittagong Univ. of Engineering and Technology, in a project *"A Fractal-Fractional Hybrid Neuron-Network Model for Neurodegenerative Disease Progression: Merging Hodgkin-Huxley Dynamics with Wilson-Cowan Mean-Field Theory".* Also, I help M.Sc. and Ph.D. students with their research work.

- **Research Assistant at CU** *From November, 2025 –March, 2026*

Worked as a Research assistant at Chittagong University under the supervision of Assistant Prof. Shahrina Akter, in a project "*The Head on Collision in Dust-Acoustic Solitary Waves in an Unmagnetized Dusty Plasma with Trapped Ion Distribution*".

- **Intern as Teaching Assistant:** *From 07 September, 2023 – 25 January, 2024*

I completed a five-month teaching internship at Katirhat Mohila Degree College in Hathazari, Chittagong, where I taught HSC first-year students and conducted Business Mathematics classes for the Management Department.

- **Associate Editor of a BCS-Math book:** *Date of Publication: November, 2022*

Book Name: ***Competitive Math Booster***. Copyright Registration No. ***27404***

Academic Credentials:

1. **Masters in Applied Mathematics**
   Institute: University of Chittagong — CGPA: 3.73/4
   Passing Year: 2022. Result Published: 2025 — *(1st class and 2nd position)*
2. **Bachelor of Science in Mathematics**
   Institute: University of Chittagong — CGPA: 3.48/4
   Passing Year: 2021; Result Published: 2023
3. **Higher Secondary Certificate (HSC)**
   Institute: Tejgaon College — GPA: 4.25/5
   Group: Science; Passing Year: *23 July, 2017*
4. **Secondary School Certificate (SSC)**
   Institute: Nakhalpara Hossein Ali High School — GPA: 4.61/5
   Group: Science; Passing Year: *30 May, 2015*

Research Experience:

1. **"Interacting Dust-Acoustic Multi-Soliton and Periodic Waves in Strongly Coupled Dusty Plasma with Ion-Drag Force"** Published in *Scientific Reports* (Springer Nature), 2026. Presented at the 24th International Mathematics Conference (24IMC 2025), Bangladesh. Co-authored with Assistant Prof. Shahrina Akter and Prof. M.G. Hafez.
2. **"Cylindrical and Spherical Propagation of Ion-Acoustic Overtaking Multi-Wave Structures in the Upper Ionosphere of Venus"** Published in *Advances in Space Research* (Elsevier), 2026. Extended the earlier Venus ionosphere work from Cartesian to cylindrical and spherical geometry, with Md. Aminul Islam, under Prof. M.G. Hafez, CUET.
3. **"Firing Dynamics of a Caputo Fractional-Order Memristive Hodgkin–Huxley Neuron Model with a Calcium Channel"** Under submission. Based on research assistantship at CUET under Prof. M.G. Hafez.
4. **"Mathematical Model for Dengue Transmission: A Comparative Study for Urban and Rural Areas in Bangladesh"** Submitted to the proceedings of the 4th International Conference on Applied Mathematics in Science and Engineering (AMSE-2025), India. *(Presented at AMSE-2025).* Collaborated with Prof. M.G. Hafez, CUET.
5. **"The Head-on Collision in Dust-Acoustic Solitary Waves in an Unmagnetized Dusty Plasma with Trapped Ion Distribution"** Currently under development. Based on research assistantship at University of Chittagong under Assistant Prof. Shahrina Akter.

Publications:

1. Islam, M.A., Tanjil, T.S. & Hafez, M.G. (2026). "**Cylindrical and Spherical Propagation of Ion-Acoustic Overtaking Multi-Wave Structures in the Upper Ionosphere of Venus**" in Advances in Space Research. DOI: https://doi.org/10.1016/j.asr.2026.09.001 (Q1 Journal, Publisher Elsevier)
2. Akter, S., Tanjil, T.S. & Hafez, M.G. (2026). "**Interacting Dust-Acoustic Multi-Soliton and Periodic Waves in Strongly Coupled Dusty Plasma with Ion-Drag Force**" in Scientific Reports, 16, 28115. Published: 19 June 2026. DOI: https://doi.org/10.1038/s41598-026-57659-z (Q1 Journal, Publisher Springer Nature)
3. Akter, S., Tanjil, T.S. & Hafez, M.G. (2026). "**Propagation and Interaction of Ion-Acoustic Single, N-Solitons, and Periodic Waves in Venus' Upper Ionosphere**" in Advances in Space Research Volume 77, Issue 4, 15 February 2026, Pages 5236-5253, https://doi.org/10.1016/j.asr.2025.12.043. (Q1 Journal, Publisher Elsevier)

Conference Presentation:

1. **AMSE-2025 International Conference, India**
   Title: Mathematical Model for Dengue Transmission: A Comparative Study for the Perspective of Urban and Rural Areas in Bangladesh.
   Date: 29 August, 2025
   Role: Presenter
2. **24IMC 2025 – International Mathematics Conference, Bangladesh**
   Title: *Interacting Dust-Acoustic Multi-Soliton and Periodic Waves in Strongly Coupled Dusty Plasma with Ion-Drag Force*
   Date: 18 December, 2025
   Role: Presenter

Technical Skills:

- Mathematical Software: **Maple, MATLAB, Mathematica.**
- Hypertext Design: **LaTeX.**
- Programming: **Python, Fortran, C++.**
- Office Software: **MS Word, MS Excel, MS PowerPoint and other Office tools.**

Additional Training:

1. **Introduction To Computer Application Packages and Unicode Bangla**.
   Conducted by *Bangladesh Computer Council*
   From 01 August to 30 October 2022.

Extra-Curricular activities:

- Member of CU Math 2021 Soccer Team. (Champion)
- Member of CU Math 2019 Cricket Team. (Champion)

Linguistic Proficiency:

- English (Fluent Working Proficiency)
- Bangla (Native Language)

References:

Dr. Md. Golam Hafez,
Professor, Department of Mathematics,
Chittagong University of Engineering and Technology

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Dr. Mohammed Aman Ullah
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Shahrina Akter
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